

Gate E1 report — run_v1.2_r1 (Paper IV v1.2)

Verdict: **PASS**. Evidence directory: 20_EVIDENCE/E1/.

E1 — Claim semantics (first pass, independent)

Inputs: EN md a501d53b... (line numbers below), literal Lean headers extracted by 20_EVIDENCE/E3/extract_headers.py (headers_*.txt). No author verdict/report was used. Labels: FACT_WITHIN_SCOPE = statement reading; the Lean column is the literal type (static; kernel confirmation is E4).

Claim map (headline and quantitative)

ID	Claim (EN line)	Scope type	Literal hypotheses/quantifiers	Lean declaration (static)	Semantic verdict
A	Thm A 1.53-61: $\exists b \forall n \geq 0 \forall \text{chordal } G: cp \leq c4 \leq M(n) + b$	all-order additive	b before n,G; chordal = cycle-chord def; $M(n) = \lfloor n(n+1)/6 \rfloor$ ($\mathbb{N}$ div)	Erdos81AllOrders.erdos81_all_orders_additive; explicit $b \leq T(h+8)$: ExplicitThreshold.erdos81_all_orders_bounded_additive	MATCH
B1	Thm B (1.1) 1.63-68	eventual exact upper	$\exists N \forall n \geq N \forall \text{chordal } G \exists Q \text{ order} \leq 4, Q \leq M(n)$	Erdos81Unconditional.erdos81_cliquePartition	MATCH
B2	Thm B (1.2) max over chordal = $M(n)$	eventual exact max, unrestricted	witness Q order ≤ 4 , size $M(n)$, minimal among all partitions	PaperTheorems.erdos81_max_eq	MATCH
C	Thm C 1.82-97: $\forall s \exists N_s \forall n \geq N_s, \text{rsd} \leq s \Rightarrow c4 \leq Q_s(n); \exists \text{attaining } G \text{ with } cp = Q_s$	fixed-defect, eventual; lower witness unrestricted	rsd def: $\forall U \forall \text{clique } R \subseteq U$ ($\emptyset$ allowed) $\exists v \in U \setminus R, \exists \text{clique } C \subseteq N_U(v), N_U(v) \leq C + s$	DefectExplicitPublication. rooted_defect_eventual/_maximum; def RootedSimplicialDefect. RootedDefectAt; defectTarget s n = targetSize(n+s) - C(s+1,2) ($\mathbb{N}$ truncation irrelevant for $n \geq N$)	MATCH (definition equivalent to [15] rsd; see E7)
C'a	Thm C' (6.12) 1.910-936	fixed s; explicit γ_s, N_s 'stab, $A_s = \max(40000(s+1)^3, 2/\epsilon_s) = 2 \cdot 10^{41}(s+1)^8$	real $\delta, 0 \leq \delta \leq \gamma_s n^2$; hyp $c4(G) \geq Q_s(n) - \delta$ ($\forall \text{order-4 } Q$); root C clique of G, $ D = s, 2 \leq C \leq H $, partition of V	SublinearResearch.fixed_defect_joint_stability_real (E32.IsDefectRoot, rootEdit, rootBaseline = $(C + D) H - C(C , 2)$)	MATCH
C'b	(6.13) $N_{\text{bad}}, E_{\text{bad}}$ for every unrestricted Q, real τ ; root before Q, τ	same-root, unrestricted partitions	$\forall Q \forall \tau \in \mathbb{R}, Q \leq Q_{s+\tau}$	same; canonical = E32. IsCanonicalPiece (CUD-H edges; C,C,H triangles); $E_{\text{bad}} = \Sigma C(K , 2)$ over noncanonical	MATCH (Lean allows any real τ , not only $\tau \geq 0$: stronger)
C'c	Cor 6.3 (6.17) optimal-size template, square-root term	fixed s	same hyp.; T admissible extremal (core size nearest to $(2(n+s)+1)/6$)	fixed_defect_exact_extremal_edit_real	MATCH; resized core not claimed clique ✓
C'd	(6.18) $cp = Q_s \Leftrightarrow c4 = Q_s \Leftrightarrow G \cong E_{s(n,k)}, k$ nearest	eventual classification	—	E32.cp_classification_of_theoremCPrimeC + FixedExplicit.fixed_defect_stability_explicit (premise supplied)	MATCH (the implication alone is conditional; the claim map says so)
6.3a	Prop 6.3a (6.20)-(6.21)	explicit family, each fixed s	$n+s=3q, 2s+2 \leq q, 1 \leq d, 2d \leq q$; every labelled optimal template	OptimalTemplateObstruction.shifted_template_witness/_sqrt_witness/no_linear_optimal_template_bound/optimal_family_nonempty	MATCH, but Lean statement is existential ($\exists G$) while the text names the explicit graph $G_{\{s,q,d\}}$; explicit form lives in the proof (OBSERVATION E1-O1)
6.1	Thm 6.1 (6.7) chordal: $M - B + m/16 + A/2 \leq \delta, d_E = m + A \leq 16\delta; \gamma = \eta 0/4$	chordal, eventual	rational δ in chordal_linear_stability_sixteen; real δ in chordal_joint_stability_real	IntegralStability / PublicationStability	MATCH (Lean also gives $2 \leq R \leq n - R $ and the 117/1825, 12687/20000 form)

6.1a	Cor 6.1a $\tau + 488$, $10\tau + 4808$, same root, unrestricted Q	chordal	text says $\tau \geq 0$; Lean any real τ	chordal_joint_stability_real	MATCH (Lean stronger in τ)
6.2	Cor 6.2 / (6.9) eventual classification cp & c4	chordal eventual	—	ExtremalClassification. chordal_extremal_classification	MATCH
6.4	Prop 6.4 (6.22) all orders incl. 0, $s \leq n$; (6.23) band $4s \leq n$	finite fractional bound; uniform integral version	certified optimum w (not a feasible dual)	certified_shifted/refined_fractional_bound_all_orders; uniform_shifted/refined_partition_bound ($\exists N \forall n s, N \leq n, s \leq n$ — threshold independent of s)	MATCH
6.5	Thm 6.5 sequences: orders $\rightarrow \infty$ (no monotonicity), $s, j = o(n_j)$; (6.24) upper excess $o(n^2)$; (6.25)-(6.26) with eventual near-extremality, rational δ_j, τ	sublinear, arbitrary order sequences	Tendsto order atTop atTop; $\forall^t$ hypothesis	sequence_arbitrary_orders_partition_bound/_stability	MATCH ((6.25) d_E expressed as missing + outside, equal to d_E(G,S,R) for a clique R)
6.6	Cor 6.6 tower $T(P(s))$, uniform $\leq S$, $s_{\max}(n)$	exact bound, uniform band	$P(s) = c.T(s+1)^{1680}$, $c.T = h + c_{\text{far}} + 841780$, $c_{\text{far}} = c_{\text{poly}}(128 \cdot 10^8 2)^{105}$	E35.theoremC_tower/_uniform/_uniform_sMax; sMaxT = Nat. findGreatest	MATCH (inverse caveat stated)
C.4	$N_{\text{far}}(\eta) \leq T(\lceil c_{\text{poly}}/\eta^{105} \rceil)$, $0 < \eta \leq 1$ rational	uniform in margin	—	E35.NfarE_le_tower_poly (cPoly = $417664 \cdot 59000^{105} + 3006$)	MATCH
3.1	Thm 3.1 $\forall \xi > 0 \exists N \forall G \forall$ rational x: $w(x) - g(P) \leq \xi n^2$ (all graphs)	uniform transfer	—	RC01Final.rc01_uniformRoundingTarget (MixedRounding. UniformRoundingTarget)	MATCH
D.2	Prop D.2 $n \equiv 4 \pmod 6$, K_n order $\leq 3 \Rightarrow > M(n)$	fixed family	—	DefectSharpPublication.order_three_insufficient	MATCH; scope limited to $s = 0/K_n$ (stated)
F.3a/b	Lemma F.3a ($ D - C - s)_+^2 \leq \text{ordNE}$; Cor F.3b	finite, all D	$\mathbb{N}$ truncated subtraction = positive part; ordered nonedges	exists_clique_additive_defect; recover_clique_from_edit (u rational ≥ 0)	MATCH
G.1	$\Sigma x.T \leq v3 + (10 + d/2)n$, chordal $K_{\{d+2\}}$ -free (annex)	fixed d, triangle (not mixed) gap	—	BoundedCliqueGap.chordal_gap_linear_cliqueFree (annex, separate)	MATCH
G.2	Prop G.1 $N0 > \exp(t^2/16)$ for the CleanupAtWith contract	contract-specific	ε rational	FarExploration. CleanupRigidVerdict.threshold_gt_exp (not re-extracted)	Not re-extracted (static text only)

Required distinctions (mandate §4 E1) — checked

- All-order additive (Thm A) vs eventual exact (Thm B, C) vs fixed-defect (C, C', 6.3, 6.3a) vs uniform-in-defect (6.4, 6.6, (C.4)) vs sublinear sequences (6.5): kept separate in text and Lean. No growing s is substituted into a fixed-s theorem (6.5 goes through SublinearEdit/[22] and RootScore).
- Unrestricted partitions: lower bounds (6.1),(6.2),(E.5),(6.20) and partition-control (6.13),(6.26),Cor 6.1a quantify over all CliquePartition (no order bound); upper bounds are order ≤ 4 .
- Actual vs resized root: C' root C is a clique of G; Cor 6.3 resized core is explicitly not claimed a clique (text l.1012; Lean gives only an admissible extremal T).
- Thresholds: (F.1) N_s^{stab} is separate from F_s ; Cor 6.6 bounds F_s only (stated l.1126).

E1 observations

- E1-O1 (OBSERVATION): Prop 6.3a / abstract "An explicit family" — Lean statements are existential; the explicit family is in the proof term, not the type.
- E1-O2 (OBSERVATION): Cor 6.1a text " $\tau \geq 0$ " and (6.13) " $\tau \geq 0$ "; Lean proves the bounds for all real τ (harmless strengthening).
- E1-O3 (MINOR, wording, l.1259): "under the stronger near-extremality hypothesis on c4" — $c_4(G) \geq Q - \delta$ is the *weaker* hypothesis (implied by $\text{cp}(G) \geq Q - \delta$ since $\text{cp} \leq c_4$); the same sentence then correctly says it applies to cp. Also reported by E7 (E7-M2).

E1 first-pass status: PASS (no semantic defect in headline statements; dynamic confirmation of elaborated types

is part of E3/E4).

PDF generated by pandoc (HTML) + PyMuPDF Story from the Markdown source; not compiled from TeX. The TeX file alongside is pandoc's standalone conversion of the same Markdown.